



# SMART MEDICINE BOX

## Developing Dynamos

TITLE: IBIP Final Presentation 2025 - 26

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## Problem Provider

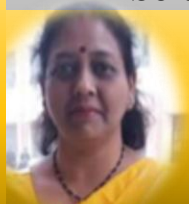
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**Dr. Shweta Anand**  
**drsa007@yahoo.com**  
**9424413044**  
**Professor (Paediatrics)**

## Solution Provider

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**Dr. Aparna Gupta**  
**aparnag@lnct.ac.in**  
**9630371294**  
**Professor(EC)**

## Solution Provider Team

0157EC231009



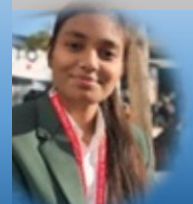
**Devansh Arjariya**  
**devansharjariya05@gmail.com**  
**6265691791**  
**Student: B.Tech V Sem (EC)**

0157EC231038



**Sunaina Pawar**  
**pawarsunaina4@gmail.com**  
**7828550071**  
**Student: B.Tech V Sem (EC)**

0157EC241004



**Aastha Vishwakarma**  
**aasthavishwakarma62@gmail.com**  
**6265618456**  
**Student: B.Tech III Sem (EC)**

0157EC231016



**Jayesh Rai**  
**raiJayesh2005@gmail.com**  
**7987795849**  
**Student: B.Tech V Sem (EC)**

LNCDMD11071



**Dr. Srashti Gupta**  
**srasht.gupta19@gmail.com**  
**7898587919**  
**Student: PG (Pediatrics)**

LNCDMD11070



**Dr. Paras Jain**  
**paras2297jain@gmail.com**  
**7415832289**  
**Student: PG (Pediatrics)**

**For chronic long term medical therapy  
Need system of –**

- ✓ Medical tracking
- ✓ Medical dispensing
- ✓ Reminder of drug



✓ **Medical dispensing**



**PROBLEM STATEMENT**

## PROBLEM STATEMENT

Need system which will improve

- ✓ Medication Adherence
- ✓ Reduce dosage error
- ✓ Monitoring for health care profession
- ✓ Ambulatory setting





## Existing Solutions – Reliability Gap

### Why Current Solutions Fail to Ensure Medication Compliance

Current medication management solutions are largely **reminder-based** and **non-enforcing**, making them unreliable in real-world clinical scenarios.

#### ➤ Commonly Used Solutions

- Basic pill organizers
- Mobile reminder applications
- Manual caregiver or nursing supervision

#### ➤ Key Limitations

- No Sound alert and user authentication .
- Reminder-based, not compliance-based
- No mechanism to confirm medicine intake
- High dependency on patient discipline
- No real-time monitoring or accountability

# PROPOSED SOLUTION

## Smart Medicine Box – Automated Medication Compliance System

An intelligent, sensor-driven medicine dispensing system designed to:

- Automate medication scheduling and access
- Ensure correct dose delivery at the prescribed time
- Confirm medicine intake through system validation
- Enable remote monitoring for caregivers and clinicians

### Connectivity Options

- Online: Wi-Fi, cloud storage, web dashboard
- Offline: GSM-based SMS and call alerts

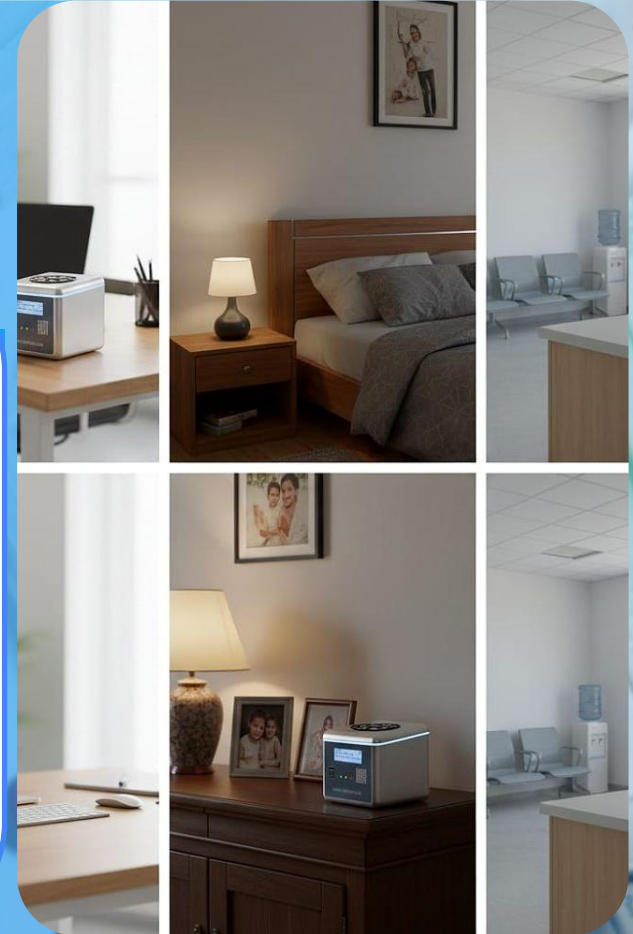


# PROPOSED SOLUTION

An IoT-enabled smart medicine box providing timely alerts and reminders, secure user identification through RFID, a cloud-connected web platform, and enhanced safety, adherence, and monitoring.

The system operates in **dual mode** — **online and offline**, ensuring reliability even in low-connectivity environments.

It's a two way communication system means user can update their data manually in box and it's also get's updated on website and vice-versa.



# MARKET COMPARISON

## Conventional Medication Reminder Systems

Fragmented & User-Dependant



- Multiple separate devices



- Manual setup and Logging ⚠️
- Limited Setup and logging ❌
- High risk of offline functionality ❌
- Strong dependence on user memory ✓



## Integrated Smart Medicine Box

Automated, Reliable & Clinically Aligned



- Single integrated solution ✓
- Fully automated medicine dispensing ✓
- Complete offline functionality ✓
- Intuitive, elderly interface ✓
- Designed for clinical and caregiver support

## Core Functional Capabilities



- RTC-driven precision scheduling using **DS3231**
- Automated compartment access via **servo motor**
- **RFID-based secure user authentication**
- **IR-based pill removal verification and medication count**



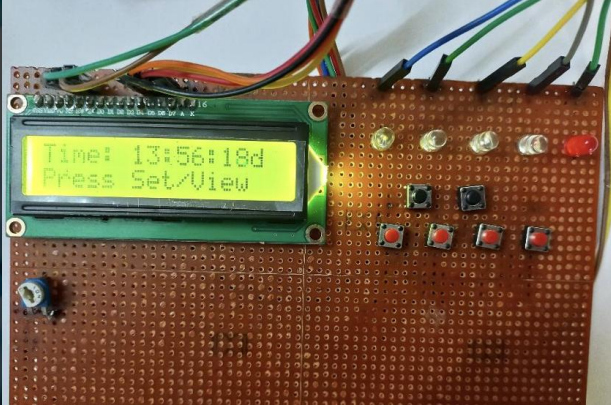
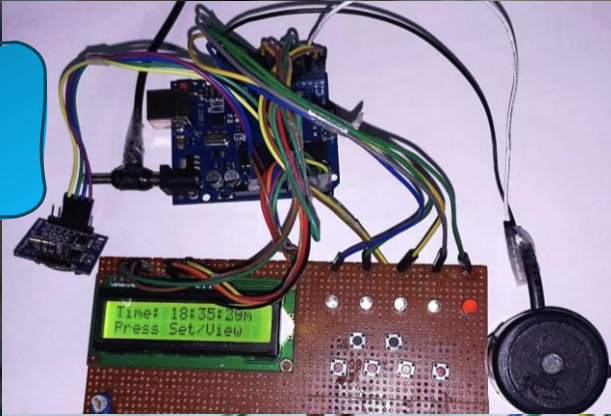
- Multi-alert system: LCD, LEDs, Speaker, and DF Player voice prompts
- Dual communication:  
Online: Wi-Fi, Firebase, Web dashboard  
Offline: GSM-based SMS and call alerts



- Local **SD card data logging**
- Emergency SOS alert mechanism

This architecture prioritizes **clinical robustness, patient safety, scalability, and affordability**

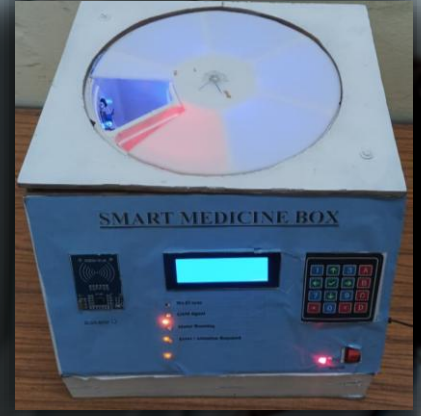
# SYSTEM IMAGES



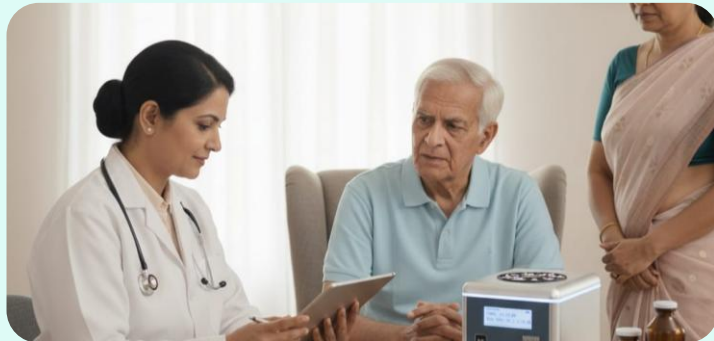
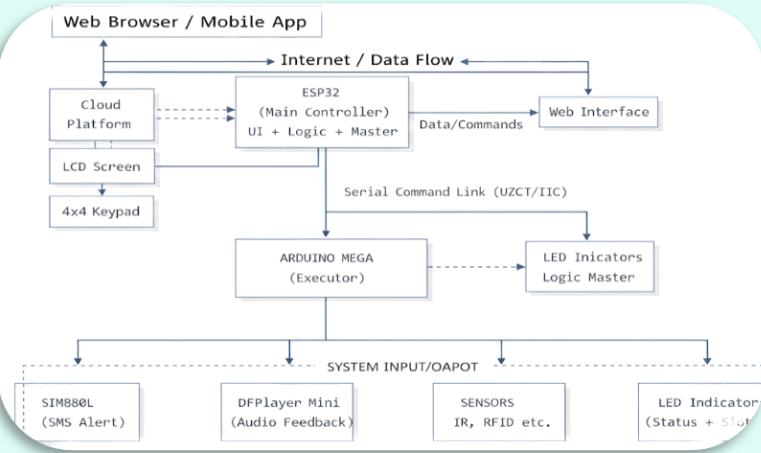
**ALPHA  
PROTOTYPE**



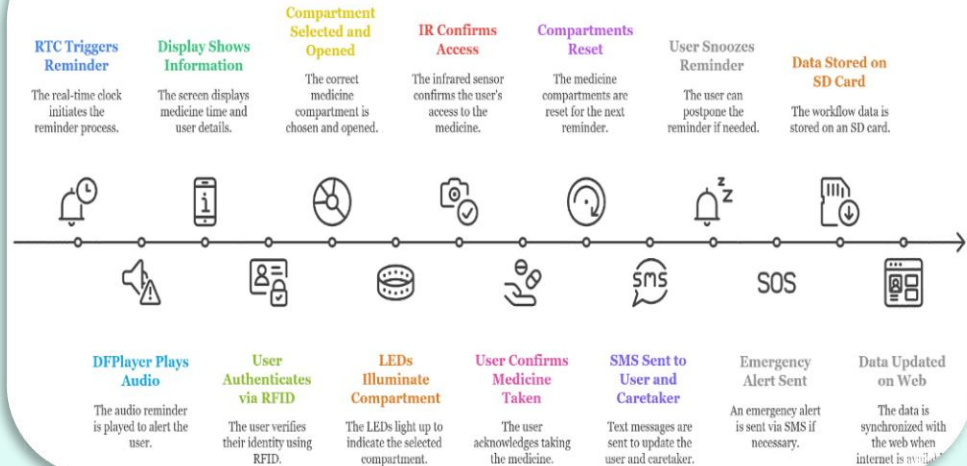
**BETA  
PROTOTYPE**



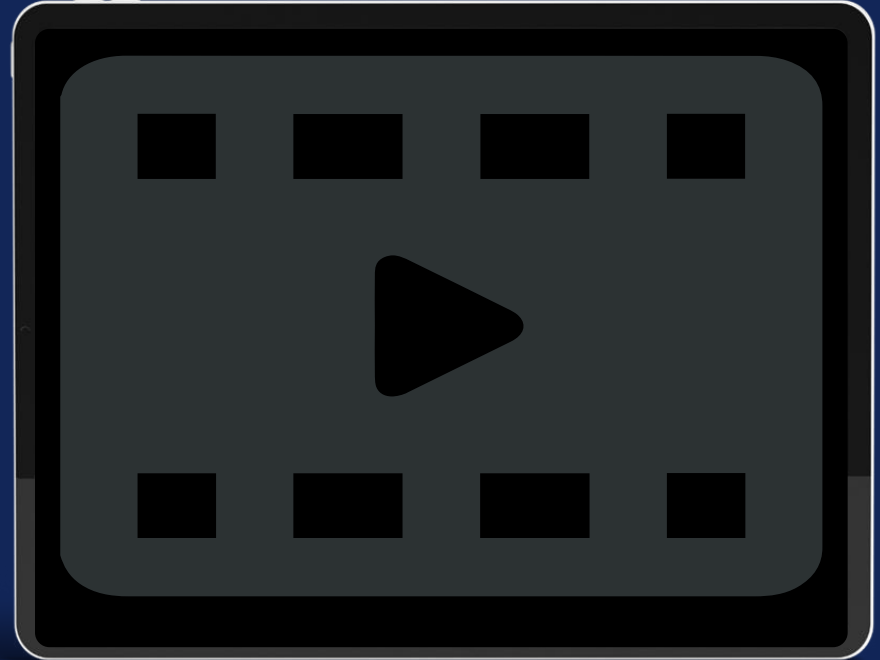
# SYSTEM ARCHITECTURE



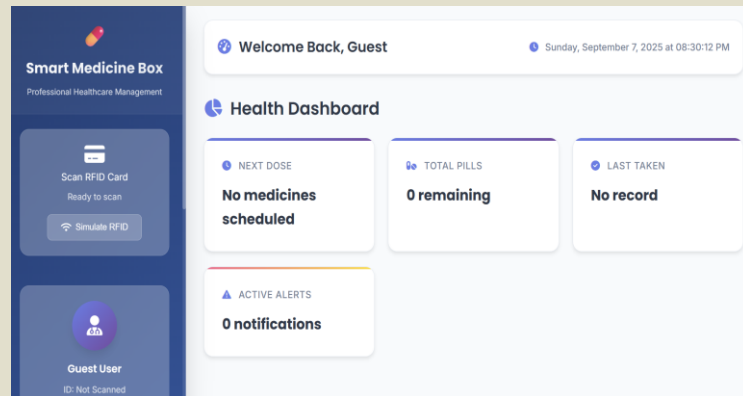
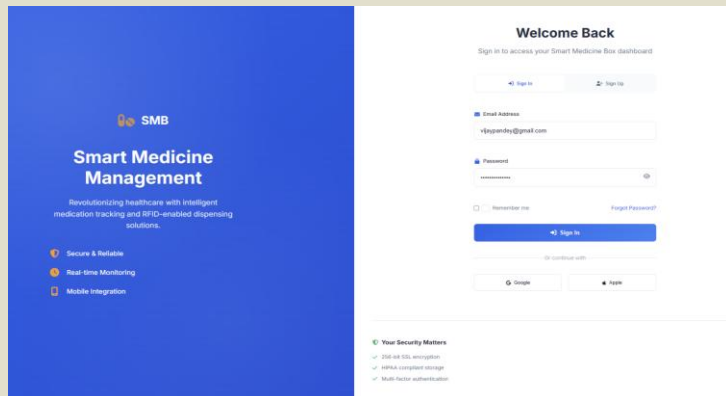
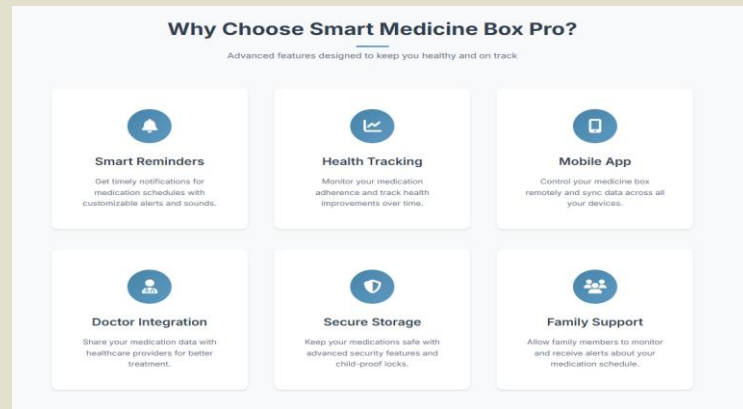
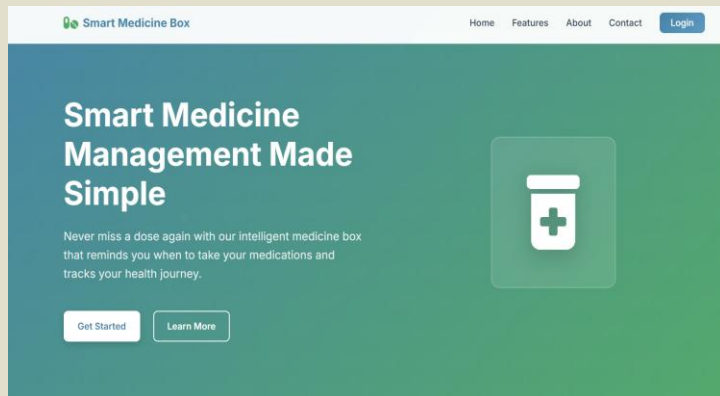
## Smart Medicine Box Workflow



**Complete  
Working  
Video of  
SMART  
MEDICINE  
BOX**



# WEBSITE PRIVIEWS



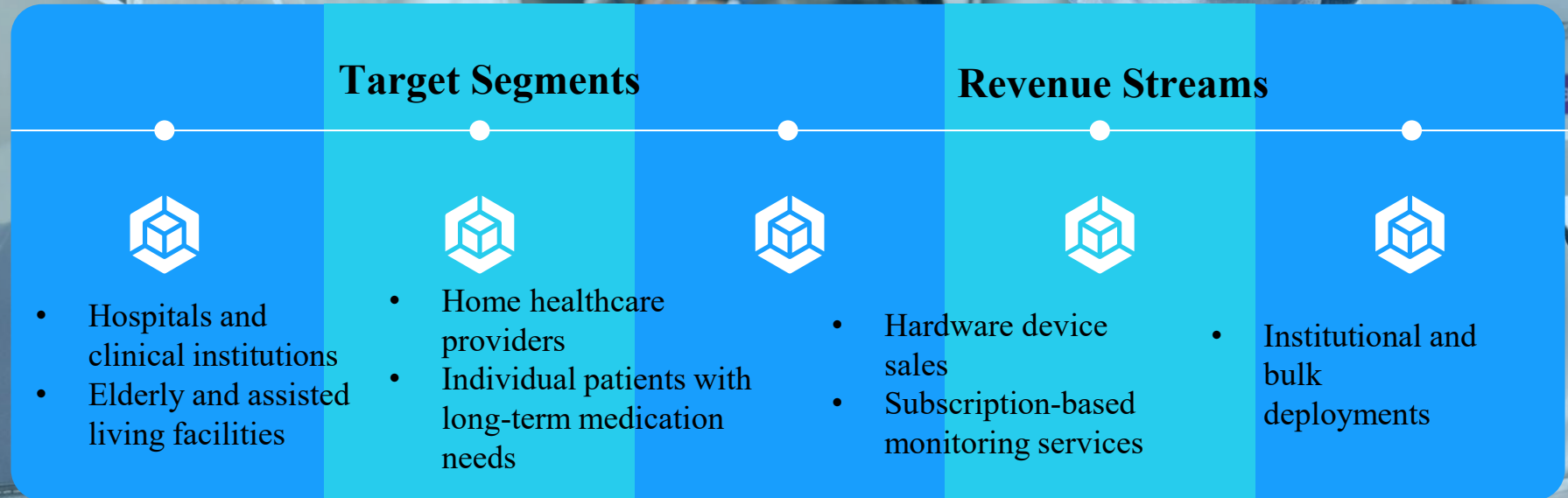
# SYSTEM CONTROL & USER INTERACTION COMPARISON

| Parameter                  | Existing Market Solutions | Smart Medicine Box                               |
|----------------------------|---------------------------|--------------------------------------------------|
| Primary Control Method     | Mobile App / Alarm-based  | <b>Web Dashboard (Full Control)</b>              |
| Secondary Control (Backup) | ✗ Not available           | <b>Offline Embedded Keypad</b>                   |
| Internet Dependency        | High                      | <b>Low (Offline-first architecture)</b>          |
| Operation Without Internet | Limited / None            | <b>Fully functional via keypad + RTC</b>         |
| User Authentication        | App login / PIN           | <b>RFID-based unique IDs</b>                     |
| Role-Based Access          | Limited                   | <b>User / Caretaker / Doctor hierarchy</b>       |
| Caregiver & Doctor Control | View-only                 | <b>Schedule, alerts, permissions</b>             |
| Emergency Handling         | App-based                 | <b>Hardware-triggered emergency + SMS</b>        |
| Alert Channels             | App notification          | <b>Audio + LED + SMS</b>                         |
| Multi-User Support         | Limited                   | <b>Multiple users with assigned compartments</b> |
| Clinical Usability         | Consumer-focused          | <b>Assisted-care &amp; clinical ready</b>        |

# DATA MANAGEMENT, RELIABILITY & SYSTEM INTELLIGENCE

| Parameter                | Existing Market Solutions | Smart Medicine Box                          |
|--------------------------|---------------------------|---------------------------------------------|
| Data Logging             | Cloud-only                | <b>SD Card + Cloud</b>                      |
| Offline Data Storage     | ✗ Not supported           | <b>Automatic SD card logging</b>            |
| Offline-to-Online Sync   | ✗ Not available           | <b>Auto-sync to web when connected</b>      |
| Dose Intake Verification | Self-reported             | <b>Hardware-confirmed intake</b>            |
| Audit Trail              | Basic app history         | <b>Time-stamped, verifiable logs</b>        |
| Compliance Analytics     | Limited                   | <b>Daily / Weekly adherence statistics</b>  |
| Missed Dose Handling     | Reminder only             | <b>Escalation to caregiver/doctor</b>       |
| System Diagnostics       | Hidden / None             | <b>User-accessible diagnostics menu</b>     |
| Fault History            | ✗ Not available           | <b>Timestamped error logs (SD + Web)</b>    |
| Power Failure Resilience | Limited                   | <b>RTC-backed continuity</b>                |
| Data Reliability         | Internet dependent        | <b>Fail-safe local storage + sync</b>       |
| Maintenance Readiness    | Low                       | <b>Preventive diagnostics &amp; reports</b> |

# BUSINESS MODEL



# SCALABILITY & LONG-TERM POTENTIAL



## SCALABILITY DIMENSIONS

- Supports multiple medicines, schedules, and patient profiles
- Can be deployed across:
  - Hospitals and clinical wards
  - Elderly care and assisted living facilities
  - Home and remote healthcare environments



## SYSTEM EXPANDABILITY

- Hardware modules can be upgraded without redesign
- Software allows centralized monitoring of multiple devices
- Suitable for both urban and rural healthcare settings



## FUTURE SCOPE

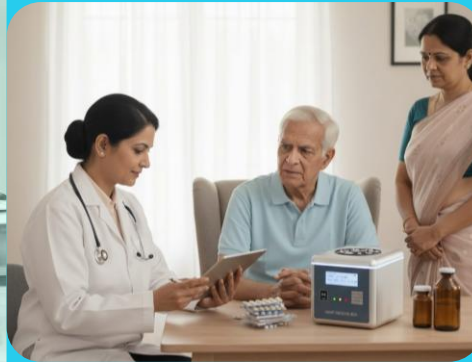
- Integration with digital health records
- Advanced compliance analytics for clinicians
- Custom configurations for institutional deployment

# CONCLUSION AND IMPACT



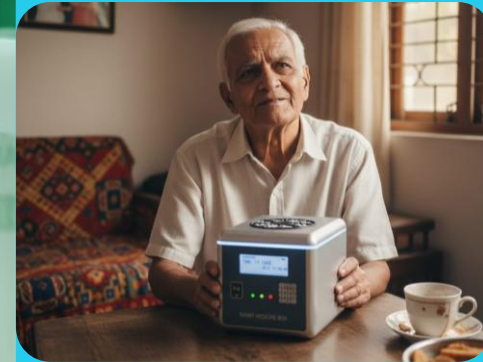
## From Reminder Systems to Medication Compliance Enforcement

The Smart Medicine Box successfully addresses the core limitations of existing medication management solutions by shifting from passive reminders to active, automated compliance monitoring.



## KEY IMPACT

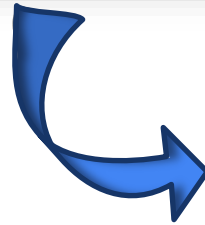
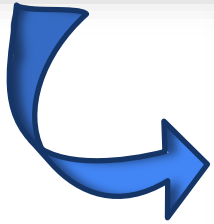
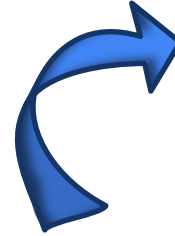
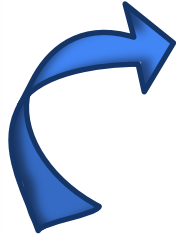
- Significantly reduces missed doses and medication errors
- Enhances patient safety through controlled access and verification
- Provides caregivers and clinicians with real-time visibility
- Minimizes dependency on memory, discipline, and manual supervision



## FINAL OUTCOME

This system represents a practical, scalable, and clinically relevant solution for improving medication adherence and strengthening modern healthcare delivery.

# Conclusion & Impact



THANK YOU

